RAJALAKSHMI ENGINEERING COLLEGE

Dinesh Karthik R 251001059-2025-IT

D2

| | |
|---|---|
| **Started on** | Tuesday, 25 August 2026, 8:45 AM |
| **State** | Finished |
| **Completed on** | Tuesday, 25 August 2026, 9:28 AM |
| **Time taken** | 42 mins 42 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

**Problem Statement:**

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

**Input Format**

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

**Output Format**

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>
int fP(int arr[], int l, int r, int x) {
    if (l >= r)
       return 0;
    int sum = arr[l] + arr[r];
    if (sum == x) {
        printf("%d\n%d", arr[l], arr[r]);
        return 1;
    }
    if (sum > x)
        return fP(arr, l, r - 1, x);
    else
        return fP(arr, l + 1, r, x);
}
int main() {
    int n, x;
    scanf("%d", &n);
    int arr[n];
    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);
    scanf("%d", &x);
    if (!fP(arr, 0, n - 1, x))
        printf("No");
    return 0;
}

```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 4<br>2<br>4<br>8<br>10<br>14 | 4<br>10 | 4<br>10 | ✔ |

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 5<br>2<br>4<br>6<br>8<br>10<br>100 | No | No | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course